

Leveraging LLMs for Video Querying

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Abstract—In the pursuit of enhancing video search capabilities, this paper delves into the application of Large Language Models (LLMs). The focal point is the utilization of the Bi-modal Transformer (BMT) model and the Vid2Seq model for video captioning, employing the UCF Crime dataset for experimentation. By employing various LLMs such as GPT3.5, GPT4, and LLAMA, the objective is to create an efficient search system that can pinpoint specific moments within videos. The paper outlines a methodology involving the generation of text-based captions using the BMT and Vid2Seq model. Furthermore, it conducts experiments using different LLMs and prompts, with evaluation metrics defined to assess system performance and event occurrence accuracy. The findings acquired provide insights into the capabilities and constraints of LLMs and prompts, illustrating their potential to enhance video captioning and transform video search functionalities. This research makes a valuable contribution by exploring LLM-based video querying techniques and their potential to improve video search efficiency.

Index Terms—Language Models, Video Caption Querying, Bi-modal Transformer, Vid2Seq, UCF Crime Dataset, Fine-tuning, Prompt Comparison

I. INTRODUCTION

Efficiently searching and pinpointing specific moments within videos is of paramount importance, particularly in the context of crime videos. This paper delves into the application of Large Language Models (LLMs) to enhance video search capabilities, with a specific focus on querying crime-related videos. The study leverages the Bi-modal Transformer (BMT) model, which is fine-tuned on a handcrafted caption dataset tailored for crime videos, resulting in a robust and contextually relevant video captioning system.

Existing video search systems often struggle with accurately identifying and retrieving relevant moments within crime videos. By harnessing the power of LLMs and fine-tuning the BMT model on a handcrafted caption dataset specifically designed for crime videos, we aim to address this challenge. This approach allows us to develop an efficient search system that can precisely identify and retrieve crucial moments related to criminal activities.

In the state of the art, the application of Large Language Models (LLMs) for video querying, particularly in the context

of crime videos, remains relatively unexplored. The capabilities of LLMs like GPT3.5, GPT4, and LLAMA have been demonstrated in various natural language processing tasks, but further investigation is needed to determine their effectiveness in enhancing video search capabilities. This research gap is addressed in this paper, which aims to explore the potential of LLMs when attached on top of the Bi-modal Transformer (BMT) and Vid2Seq models, for efficient and precise video querying. The captions generated by the BMT and Vid2Seq models are queried using these LLMs, enabling improved video search capabilities in the domain of crime videos.

The proposed solution involves fine-tuning the BMT model on a carefully curated caption dataset consisting of crime videos. This dataset encompasses various criminal activities, allowing the model to capture the nuances and specificities of crime-related context. By training the BMT model on this dataset, its ability to generate accurate and relevant captions tailored to the crime domain is enhanced.

To assess the system's efficiency and gauge the precision of event detection in crime videos, trials are carried out utilizing various LLMs, such as GPT3.5, GPT4, and LLAMA. By comparing their outputs and analyzing the impact of various prompts, insights are gained into the strengths and limitations of LLM-based video querying techniques in the crime domain.

The major contributions of this paper are as follows:

- The performance of Large Language Models in the domain of Video Querying and the accuracy of the state-of-the-art LLM models in pinpointing the occurrence of an event or activity in a long-form video are studied.
- A handcrafted captioning dataset is introduced for finetuning the BMT model, and performance gains are reported by utilizing this finetuned BMT model for video querying.
- The results returned by different LLMs on a specific task of querying crime videos are compared, and the performance is evaluated by defining a custom metric.

The paper is organized as follows: In Section II, we review the existing literature on video search, captioning using LLMs, and the specific challenges associated with crime videos. Section III provides an in-depth description of our methodology, emphasizing the fine-tuning process of the BMT model on

the handcrafted caption dataset for crime videos. In Section IV, we outline the evaluation metrics utilized to assess the system’s performance and the dataset and experiments used to achieve the corresponding results. Section V presents the comprehensive results obtained from our experiments, highlighting the impact of different LLMs and prompts on the system’s performance in querying crime videos. Finally, in Section VI, we discuss the implications of our findings, the potential applications of LLM-based video querying in crime investigation, and future research directions.

II. LITERATURE SURVEY

Video captioning models, as referenced in [4], are deep learning architectures specifically created to autonomously produce written explanations or captions for video content. These models are intended to narrow the divide between visual content and natural language comprehension by scrutinizing the visual content within videos and producing corresponding textual descriptions. Typically, video captioning models follow a two-step methodology. Initially, a video encoder is used to analyze the input video frames or clips, extracting visual features that encapsulate both the temporal and spatial aspects of the video. [13], [20] These features encode the visual context necessary for generating accurate captions. [5] In the second stage, a language decoder takes the encoded video features and generates the textual captions. [21] This decoder’s choice can depend on the particular architecture employed, which may include recurrent neural networks (RNNs), convolutional neural networks (CNNs), or transformer models. The decoder generates the captions by sequentially predicting words or phrases [10], conditioned on the visual features and previously generated text. [7]. To train video captioning models, large-scale datasets containing videos paired with human-annotated captions are essential. These datasets provide the necessary supervision for the models to learn the mapping between visual content and textual descriptions. The training procedure entails fine-tuning the model’s parameters to minimize the disparity between the generated captions and the reference captions.

Bi-modal Transformers [1], [16], a vision-language model based on transformers, have garnered significant attention and demonstrated remarkable strengths in the field of video querying. These models effectively capture complex relationships within and between visual and textual representations, enabling robust performance in tasks like visual question answering (VQA) and bi-directional image and text generation [15]. The integration of the Contrastive Language Image Pretraining (CLIP) network facilitates the seamless embedding of image patches and question words, while attention mechanisms adeptly capture dependencies. The averaging of predictions from classifiers mounted on top of contextual representations yields reliable final answers [12]. Moreover, the adoption of a unified multimodal framework based on the Transformer architecture showcases the flexibility and adaptability of Bi-modal Transformers in generating coherent sequences of images and text. Bi-modal Transformers exhibit remarkable strengths in video querying, capturing intricate relationships within visual and textual representations [2]. Nonetheless, it is crucial to recognize their limitations, such as potential biases stemming

from training data, computational complexity, and performance variations across domains and datasets. To overcome these challenges and bolster the applicability and reliability of Bi-modal Transformers in video querying, addressing these weaknesses necessitates meticulous attention to fair training data curation, optimization of computational efficiency, and robust generalization across diverse contexts. By addressing these factors, the potential of Bi-modal Transformers can be fully harnessed for enhanced video querying capabilities.

Large Language Models (LLMs)[3] represent sophisticated deep learning systems that have undergone extensive training on extensive text datasets to comprehend and produce language that closely resembles human communication[18]. These models harness advanced neural network structures, such as transformers, to analyze and generate text by applying the patterns and structures acquired from their training data. The introduction of Large Language Models (LLMs) has brought about a transformative impact on the realm of language modeling, capitalizing on advanced deep learning methodologies and transformer-based architectures to grasp and generate language that closely mirrors human expression. Notable instances of LLMs encompass models like GPT 3.5, GPT4 [11], [14], and LLaMA [9]. These models showcase substantial improvements in language understanding and generation capabilities, surpassing their predecessors and achieving competitive performance on various natural language processing benchmarks. By training on extensive and diverse text corpora, these LLMs capture intricate patterns and structures, enabling them to generate contextually rich and coherent text [8]. The incorporation of self-attention mechanisms and multi-layered feed-forward networks further enhances their ability to process and comprehend language [17], [19]. These advancements in LLMs pave the way for a wide range of applications in areas such as information retrieval, text generation, and natural language understanding, driving the progress of language processing technologies. Large Language Models (LLMs), including GPT 3.5, GPT4, and LLaMA, represent notable strides in language understanding and generation. These models excel in capturing intricate linguistic patterns and generating coherent text through the utilization of transformer architectures and extensive training data. The strengths of LLMs lie in their ability to comprehend language nuances and produce high-quality outputs. However, it is crucial to address potential biases in the training data, which can influence the generated results, and to tackle the computational complexity and resource requirements associated with LLMs. Despite these challenges, the transformative potential of LLMs in domains such as natural language processing, content generation, and information retrieval is immense. By mitigating biases, optimizing computational efficiency, and establishing rigorous evaluation frameworks, LLMs can reshape the landscape of language processing and contribute to significant advancements in artificial intelligence research and applications.

The UCF-Crime dataset comprises an extensive collection of 128 hours’ worth of video footage. It encompasses 1900 unedited, real-life surveillance videos, featuring 13 common real-world abnormalities. These anomalies encompass activi-

ties such as Abuse, Arrest, Arson, Assault, Road Accident, Burglary, Explosion, Fighting, Robbery, Shooting, Stealing, Shoplifting, and Vandalism. The selection of these anomalies is based on their potential to pose significant threats to public safety. Researchers have extensively utilized this dataset to develop and evaluate algorithms for crime detection, anomaly detection, and activity recognition tasks. The strengths of the UCF Crime Dataset lie in its realistic and challenging scenarios, enabling the development of robust and accurate video analysis models. However, it is important to acknowledge certain limitations, such as the relatively limited size of the dataset and potential biases in the types of crimes represented. Addressing these weaknesses through data augmentation techniques, expanding the dataset size, and ensuring diverse and representative crime samples will enhance the dataset's utility and generalizability. The UCF Crime Dataset serves as a valuable benchmark for evaluating and advancing video analytics techniques in the domain of crime detection, contributing to the development of more effective and reliable surveillance systems.

III. RESEARCH CONTRIBUTION

The performance and capabilities of Large Language Models (LLMs) are examined in this paper through exploration of several research questions. Firstly, the effectiveness of combining LLMs with other transformer architectures is investigated to tackle more complex tasks seamlessly, eliminating the need for domain adaptation. Secondly, the influence of incorporating multiple layers of modal data conversion on the model's performance in the context of generating captions for crime videos is examined. Lastly, the underlying factors that contribute to the performance disparities among various LLMs when it comes to querying lengthy videos and specifically detecting crime-related activities are sought to be identified.

The research work focuses on the utilization of two video captioning models, namely the BMT-based Dense Video Captioning model and the Vid2Seq model. The BMT model's architecture comprises the following components: an initial implementation involves the utilization of a hand-annotated dataset of captions for the UCF-Crime dataset, as outlined in Section IV-A. Subsequently, the BMT captioning model is fine-tuned, enabling the achievement of exceptional performance in video description—an essential aspect for our architecture, as the BMT model's output is subsequently processed by the LLM. One of the key advantages of the BMT model is its reliance on I3D features for extracting visual embeddings from the given video. Additionally, it incorporates VGGish features for generating audio embeddings from any audio content present in the video. These combined visual and audio embeddings are then passed through the Bi-Modal encoder layer, which employs self-attention mechanisms to effectively incorporate both visual and audio information within the model. Consequently, the model can better utilize the information encoded in the video, enabling the generation of relevant tokens or embeddings. These embeddings are subsequently forwarded to the Bi-Modal decoder layer for generating captions. Before passing the embeddings to the decoder block, they undergo processing by a proposal generator block. This block ensures

that only relevant frames within the video are utilized, and simultaneously associates the timestamp of the event or clip with the embeddings prior to their transmission to the decoder block. The decoder block itself is a captioning model that has been fine-tuned specifically to generate text related to crime activities.

The decoder block generates captions for each region of the proposal, which are then concatenated with the corresponding timestamp and accumulated using the caption accumulator module. However, in the case of long form surveillance videos, the length of the final text generated by the video may exceed the maximum context window supported by most Large Language Models (LLMs). To address this issue, a Recursive LLM Strategy Block is employed, which decomposes the caption summary of the video into multiple chunks. Each chunk is processed separately, combined with the query from the LLM, and stored in a temporary data store. This process is repeated for all the chunks of the caption summary, iteratively handling the results obtained from the previous iteration until a point is reached where all the information can fit within the context window of the LLM. It is important to note that there may be multiple occurrences of the same event within the entire window. Thanks to the recursive strategy, the system is able to track these different occurrences in various parts of the video, enhancing the relevance of the final result generated by the LLM. The architecture of this approach is illustrated in Figure 1.

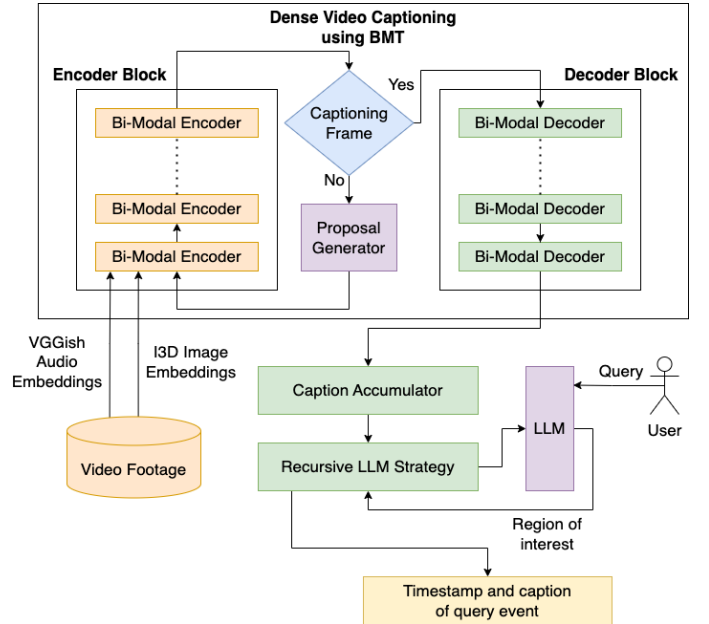


Fig. 1. System Architecture of BMT-based captioning model that uses a user-defined query for timestamp retrieval based on captions

The second video captioning model, Vid2Seq, follows a similar strategy. However, the key difference lies in the architecture of the captioning model itself. In the case of Vid2Seq, the model leverages transcribed text obtained from the audio of a given video sample, along with visual embeddings extracted from different frames of the video. The audio transcription is performed using the Google Cloud API. The architecture

of Vid2Seq consists of individual encoder blocks that process the textual information and the video frames as separate inputs. These inputs are passed through their respective encoder blocks, and the resulting representations are concatenated. The concatenated representation is then forwarded to the decoder block, which generates captions for different regions of the video. Figure 2 illustrates the architecture of Vid2Seq, showcasing the flow of information from the audio transcription and visual embeddings to the encoder blocks and finally to the decoder block for caption generation.

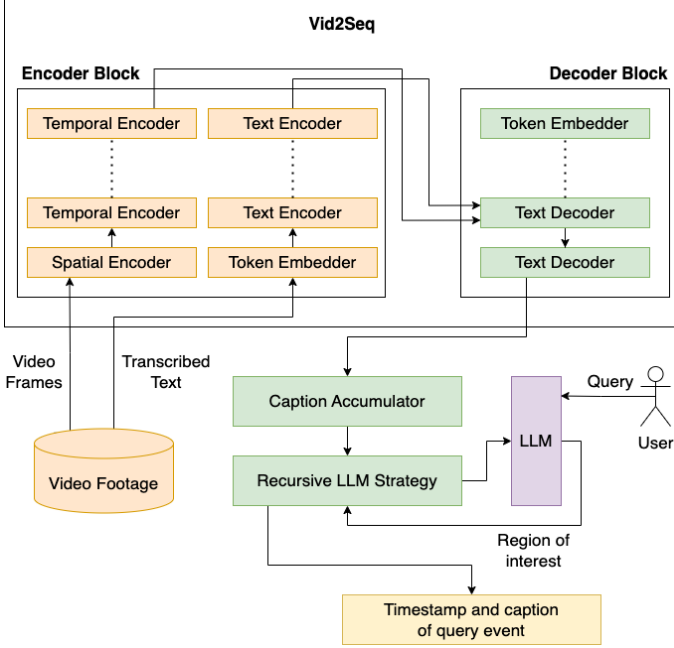


Fig. 2. System Architecture using Vid2Seq that uses a user-defined query for timestamp retrieval based on captions

IV. EXPERIMENTAL SETUP

In this section we discuss about the experimental setup of our project. Sub-section IV-A discusses the nature of the dataset that we use, IV-B outlines the training process giving specific details about the training procedure, IV-C explains the process of integrating an LLM with the pipeline.

A. Dataset

To enhance the quality of our dataset and enable precise analysis, we carried out a careful process of hand annotating 50 videos from the UCF Crime dataset. The goal was to provide clear descriptions of the contents within each video frame. To ensure accuracy and avoid any confusion, we involved three annotators in the process. We randomly selected frames from the dataset and asked each annotator to describe what they saw in a single sentence. By having multiple annotators, we aimed to capture different perspectives and reduce any potential biases. If there were significant differences in the descriptions provided by the annotators, we included a fourth annotator to help reach a consensus. This step was important to maintain consistency and address any potential disagreements. Our annotation guidelines emphasized brevity

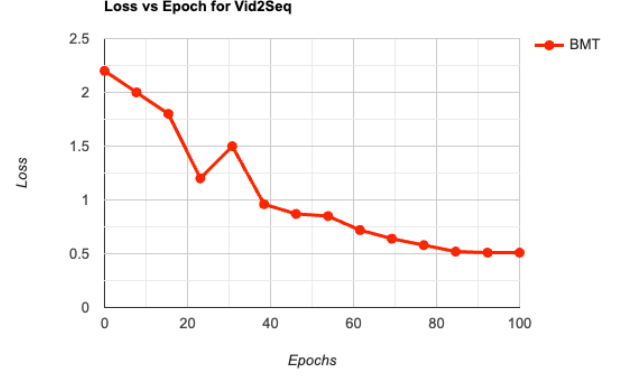


Fig. 3. Loss Vs Epoch Graph for BMT

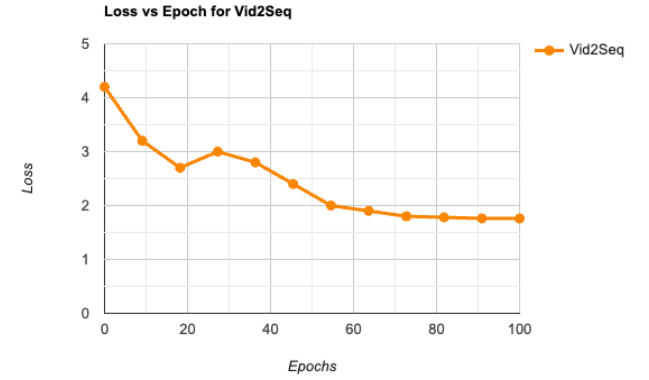


Fig. 4. Loss Vs Epoch Graph for Vid2Seq

and relevance, encouraging annotators to focus on important visual elements, objects, actions, and context within each frame. The annotations we obtained through this thorough process serve as reliable reference points for evaluating our video querying system. They also enable us to compare and assess the performance of different LLMs accurately. By involving multiple annotators and implementing a consensus mechanism, we ensured the reliability and accuracy of our hand annotations. These annotations form a solid foundation for our subsequent research and analysis tasks.

B. Model Training details

The BMT and Vid2Seq model were trained utilizing a handcrafted crime video captioning dataset, which comprises a diverse collection of crime-related videos. The training process involved several key steps. First, the captioning model was initialized with pre-trained weights from a state-of-the-art image captioning model, allowing the effective capture of high-level visual features. Next, the model was fine-tuned using the crime video captioning dataset. The dataset consists of paired image-caption samples, where each image corresponds to a frame extracted from crime-related videos, and the caption describes the activities or events in the video segment. During the training process, the model was trained with a batch size of 32, and we employed the Adam optimizer with a learning rate

of 0.001. To improve the model’s capacity for generalization, we also incorporated data augmentation techniques, including random cropping, flipping, and rotation, throughout the training process. This augmentation helps the model learn robust representations by exposing it to a wider range of variations in the data. The training process lasted for 100 epochs, with each epoch iterating over the entire dataset. The model underwent training on a GPU cluster consisting of four NVIDIA V100 GPUs. Each epoch took roughly 1.5 hours, culminating in a total training duration of 30 hours. The objective of fine-tuning the captioning model on the meticulously crafted crime video captioning dataset was to empower the model to produce precise and detailed captions explicitly designed for content related to criminal activities in videos. The details of the training process and the use of a carefully curated dataset contribute to the model’s ability to understand and describe various crime scenarios depicted in videos. Furthermore, for the finetuning of the Vid2Seq model, the Google Cloud API was utilized to generate transcriptions for audio present in the videos. These transcriptions were passed as input to the model during training.

C. LLM queries

During the video querying process, users input their queries in natural language to specify the desired events, actions, or descriptions they are searching for within the videos. These user queries are then used as input for the language models, which interpret the user’s intentions and generate relevant responses accordingly.

For instance, a query like ”Identify the exact moment when a car accident occurs in the video”. The underlying model responsible for generating the captions outputs it in the following format

$$L = [\{ST_1, ET_1, SEN_1\}, \{ET_1, ET_2, SEN_2\}, \dots, \{ET_{n-1}, ET_n, SEN_n\}] \quad (1)$$

where ST, ET and SEN is the start time, end time and the sentence generated for the interval

$$[S_k, E_k]$$

It is often beneficial to create queries that are more generic in nature to increase the chances of finding relevant occurrences within the video. For instance, a query like ”Locate the timestamp of a crowd gathering in the video” allows the model to identify any instance where a crowd gathers, regardless of the specific context or location. By carefully crafting queries with appropriate specificity, context, and generality, users can effectively communicate their requirements to the language models. This approach maximizes the chances of obtaining accurate and relevant responses.

If the prompt is generic, the output is broken down in subsequent hashes -

$$L_1, L_2, \dots, L_k$$

These hashes are then re-processed by the LLM and the number of hashes are pruned down to half and so on, producing better results.

Upon receiving the user queries, the language models leverage their language understanding capabilities to interpret and process the queries. They analyze the textual input, extract the relevant information, and generate responses in the form of timestamps that indicate where the specified events might have occurred in the given video.

V. RESULTS AND DISCUSSION

Let us define the set of segments that we are looking for as $[(l_1, r_1), (l_2, r_2), \dots, (l_m, r_m)]$. Here l_i denotes the start point of the i th segment and r_i denotes the end point. Let us suppose that the model returns a set of predictions for a given query as follows : $[(l_{pred1}, r_{pred1}), (l_{pred2}, r_{pred2}), \dots, (l_{predm}, r_{predm})]$. We define an exact overlap as the predictions where the predicted values are an exact overlap of the original expected segments.

TABLE I
ACCURACY MEASURE FOR CRIME VIDEOS

Models	Exact Overlap (%age of segments)	Overlap with 1 min temporal deviation (%age of segments)	Overlap with 2 min temporal deviation (%age of segments)
GPT 3.5	53	72	80
GPT 4	56	77	85
LLaMA	52	73	78

$$Acc_{t=0} = \frac{\mathbb{1}|l_i - l_{predi}| + |r_i - r_{predi}| = 0}{m} \quad (2)$$

$$Acc_{t=1} = \frac{\mathbb{1}|l_i - l_{predi}| + |r_i - r_{predi}| \leq 2}{m} \quad (3)$$

$$Acc_{t=2} = \frac{\mathbb{1}|l_i - l_{predi}| + |r_i - r_{predi}| \leq 4}{m} \quad (4)$$

$\mathbb{1}$ denotes the indicator function and $Acc_{t=i}$ denotes the accuracy of the model with an allowed deviation of i minutes. For e.g. if we are expecting an exact overlap between our original expected clip and the predicted clip, a result of {start: 2:00, end: 5:00} would be considered as a wrong prediction for the clip {start: 2:00, end: 4:00}. However, this prediction will be considered valid if we allow a temporal deviation of 1 min from the start or end of the clip.

Upon evaluation, we observed differences in the responses generated by the models. GPT3.5 exhibited a strong performance, providing accurate and relevant descriptions of the events in the videos. GPT4 showcased further improvements, surpassing GPT3.5 in terms of precision and contextual understanding. LLAMA, being a specialized LLM, showed a competitive performance, particularly in understanding nuances and delivering accurate timestamps for the desired events. We tabulate our results using this evaluation metric in Table I.

The variations in the responses can be attributed to the differences in model architectures, training data, and prompt engineering approaches. GPT4 with its larger parameter sizes and extensive training on diverse datasets, exhibited enhanced performance compared to GPT3.5. LLAMA, designed specifically for video captioning tasks, leveraged its specialized training to provide accurate temporal information.

Overall, GPT4 demonstrated superior performance compared to GPT3.5 and LLAMA, primarily due to its larger model

sizes, comprehensive training, and enhanced adaptability to different domains. The improvements in contextual understanding and precision offered by GPT4 contributed to its superior performance in accurately pinpointing the occurrence of events in the queried videos.

VI. CONCLUSION

In this paper we explore an amalgamation of Transformer based models used for dense video captioning with LLMs and observe that LLMs like GPT-4 are capable of querying over video content if they are provided with sufficient textual context for the given video. The large model size of GPT-4 gives it an advantage over other models while evaluating on crime video footages and also note that Bi-Modal Transformers are better at generating relevant captions for long-form videos as compared to the Vid2Seq Model. We also report an increase in performance when both of these models are specifically finetuned on the crime video captioning dataset and note that it is feasible to deploy a commercialized system that leverages this architecture.

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